Customer Shopping Behavior Analysis

End-to-End Retail Analytics Project using Python, MySQL Workbench & Power BI

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Tools & Technologies: Python (Pandas, NumPy, Matplotlib, Seaborn), MySQL Workbench, Power BI, Microsoft Excel

1. Executive Summary

Retail businesses generate large volumes of transactional data that can be leveraged to understand customer purchasing behavior and improve business performance. This project analyzes customer shopping behavior using a dataset of approximately **3,900 customer transactions** to identify purchasing trends, customer segments, product performance, and revenue patterns.

An end-to-end analytics workflow was implemented using **Python** for data cleaning and exploratory data analysis (EDA), **MySQL Workbench** for business-oriented SQL analysis, and **Power BI** for developing an interactive dashboard that communicates key business insights.

2. Business Objective

The objective of this project was to transform raw retail transaction data into meaningful business insights by answering the following questions:

- Which customer groups contribute the highest revenue?
- Which product categories generate the most sales?
- How does subscription status affect purchasing behavior?
- Which products receive the highest customer ratings?
- What impact do discounts have on customer purchases?
- Which age groups contribute the highest revenue?

3. Dataset Overview

The dataset contains approximately **3,900 retail transactions** and **18 customer and purchase-related attributes**.

Key Variables

Customer Information

- Customer ID
- Age
- Gender
- Location
- Subscription Status

Purchase Information

- Category
- Item Purchased
- Purchase Amount
- Season
- Shipping Type
- Discount Applied
- Review Rating
- Previous Purchases

During the preprocessing stage, missing values were identified in the Review Rating column and handled appropriately before further analysis.

4. Methodology

The project followed a structured analytics workflow consisting of four stages.

Stage 1: Data Cleaning & Exploratory Data Analysis (Python)

The dataset was imported into Python using Pandas for preprocessing and exploration.

The following operations were performed:

- Imported the dataset
- Examined data types and summary statistics
- Identified missing values
- Imputed missing review ratings using category-wise median values
- Renamed columns using snake_case naming convention
- Removed redundant variables
- Created new analytical features including Age Group and Purchase Frequency
- Exported the cleaned dataset for SQL analysis

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Sample Python Code

None

```
import pandas as pd

df = pd.read_csv("customer_shopping_data.csv")

df.info()

df.describe()

df.isnull().sum()
```

None

```
df['review_rating'] = (
    df.groupby('category')['review_rating']
      .transform(lambda x: x.fillna(x.median()))
)
```

None

```
df.columns = (
    df.columns
      .str.lower()
      .str.replace(" ","_")
)
```

None

```
frequency_mapping = {
    'Weekly':7,
    'Monthly':30,
    'Quarterly':90,
    'Annually':365
}
```

```
df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping)
```

5. Business Analysis using SQL (MySQL Workbench)

The cleaned dataset was imported into **MySQL Workbench**, where SQL queries were used to answer business-focused analytical questions.

The analysis included:

- Revenue generated by gender
- Highest-rated products
- Customer segmentation
- Subscriber versus non-subscriber comparison
- Revenue by age group
- Shipping type analysis
- Discount dependency
- Product rankings within categories
- Repeat customer analysis

Sample SQL Query

Revenue by Gender

```
None
SELECT gender,
SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

Top Rated Products

```
None
SELECT item_purchased,
ROUND(AVG(review_rating),2) AS average_product_rating
FROM customer
```

```
GROUP BY item_purchased
ORDER BY average_product_rating DESC
LIMIT 5;
```

Customer Segmentation

```
None
SELECT
CASE
WHEN previous_purchases <=1 THEN 'New'
WHEN previous_purchases BETWEEN 2 AND 5 THEN 'Returning'
ELSE 'Loyal'
END AS customer_segment,
COUNT(*) AS customers
FROM customer
GROUP BY customer_segment;
```

6. Dashboard Development (Power BI)

The final stage involved developing an interactive Power BI dashboard to visualize the findings and facilitate business decision-making.

The dashboard includes:

KPI Cards

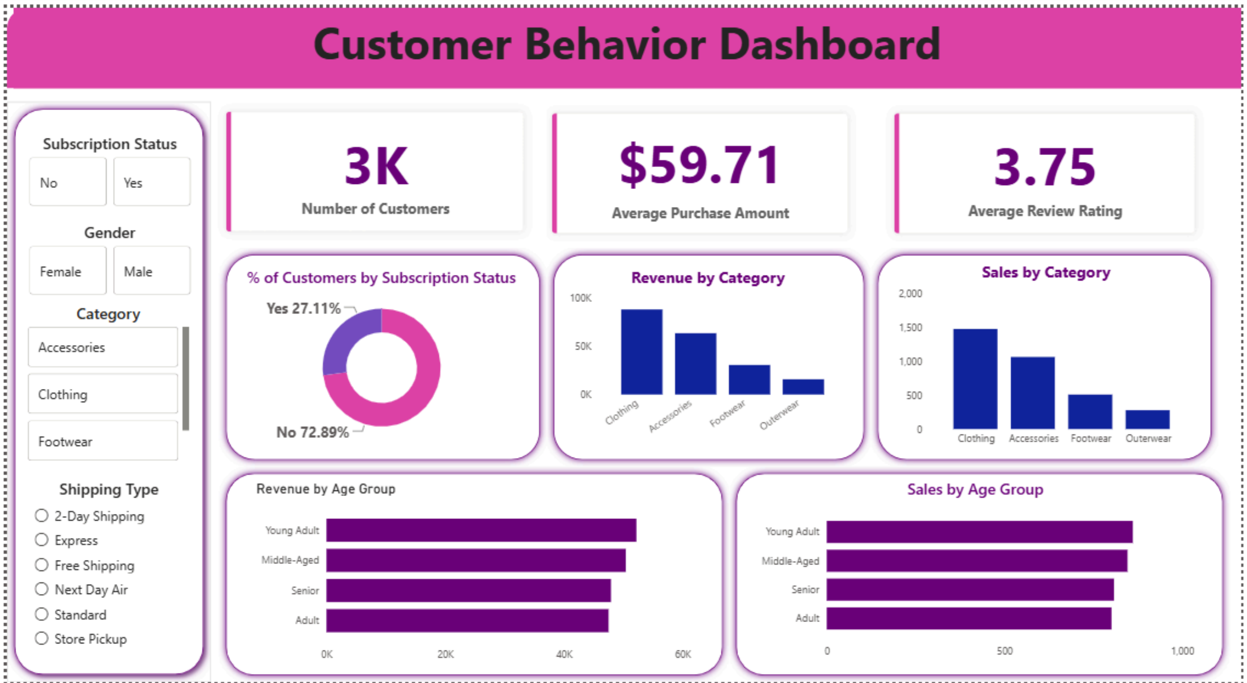
- Total Customers
- Average Purchase Amount
- Average Review Rating

Interactive Filters

- Subscription Status
- Gender
- Category
- Shipping Type

Visualizations

- Revenue by Category
- Sales by Category
- Revenue by Age Group
- Sales by Age Group
- Subscription Distribution



7. Key Findings

The analysis revealed several valuable business insights:

- Clothing generated the highest revenue among all product categories.
- Young Adults contributed the largest share of revenue.
- Loyal customers represented the largest customer segment.
- Non-subscribers accounted for the majority of customers.
- Products with higher review ratings demonstrated stronger customer satisfaction.
- Certain products relied heavily on discounts, indicating opportunities to optimize pricing strategies.

8. Business Recommendations

Based on the findings, the following recommendations are proposed:

Improve Customer Loyalty

Strengthen subscription benefits and customer loyalty programs to increase customer retention.

Optimize Discount Strategy

Reduce dependency on discounts for products with consistently high demand while maintaining promotional campaigns for slower-moving products.

Target High-Value Customers

Develop personalized marketing campaigns for age groups contributing the highest revenue.

Promote High-Rated Products

Highlight top-rated products in promotional campaigns to improve customer engagement and sales.

Enhance Business Monitoring

Continue using interactive dashboards to monitor customer behavior, revenue trends, and purchasing patterns in real time.

9. Skills Demonstrated

Programming

- Python
- Pandas
- NumPy

Database

- MySQL Workbench
- SQL
- Aggregate Functions
- CASE Statements
- Window Functions
- Common Table Expressions (CTEs)

Data Visualization

- Power BI

- Interactive Dashboards
- KPI Development
- Data Storytelling

Analytics

- Data Cleaning
- Exploratory Data Analysis
- Feature Engineering
- Customer Segmentation
- Business Intelligence

10. Conclusion

This project demonstrates an end-to-end data analytics workflow, showcasing the ability to transform raw retail transaction data into actionable business insights. By integrating Python for data preparation, MySQL Workbench for analytical querying, and Power BI for interactive visualization, the project highlights practical skills in data analysis, SQL, dashboard development, and business storytelling. The insights generated can support retail organizations in improving customer engagement, optimizing marketing strategies, and making informed business decisions.